Real Closedness of Hahn and Puiseux Series Fields

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Abstract

Let k be a real closed ordered field and let Γ be a divisible ordered abelian group. We prove that the Hahn field $k((\Gamma))$ is real closed, and that its bounded-denominator subfield with $\Gamma = \mathbb{Q}$ is real closed as well. The first proof combines valuation-theoretic normalization with transfinite Newton corrections. At a limit stage the corrections are summed over the whole ordinal initial segment, and a set-theoretic size argument rules out an indefinitely increasing sequence of exponents of the added monomials. The second proof keeps the corrections inside one rational denominator lattice. A correction whose selected root retains the current multiplicity cannot leave that lattice, so an infinite sequence of such exponents is unbounded and its Hahn sum is a Puiseux root.

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Proof overview

The proof is organized as follows. The ordered-field criterion reduces real closedness to square roots of nonnegative elements and roots of odd-degree polynomials. For Hahn series, the odd-degree problem is converted to coefficient conditions over the valuation ring. Multiplicity one is solved by ordinal recursion. At larger odd multiplicity, a Newton correction either reduces the selected root multiplicity or preserves it; the latter case is continued over complete ordinal initial segments; a sufficiently long continuation would give an impossible injection into Γ. For Puiseux series, the root construction is repeated while preserving one common denominator lattice, using Newton iteration for a simple root over the corresponding power-series ring.

1 Main statements and the ordered-field criterion

An ordered field is called *real closed* if it has no proper algebraic extension to which its order extends; equivalently, adjoining a square root of -1 gives an algebraically closed field. Let k be a real closed ordered field and let Γ be a divisible linearly ordered additive group. Here divisible means that, for every $\gamma \in \Gamma$ and every integer $n \geq 1$, there is a (necessarily unique) $\delta \in \Gamma$ with $n\delta = \gamma$. A Hahn series over k with exponents in Γ is a formal sum

$$x = \sum_{\gamma \in \Gamma} a_\gamma t^\gamma$$

whose support

$$\operatorname{supp}(x) = \{\gamma \in \Gamma : a_\gamma \neq 0\}$$

is well ordered, meaning that every nonempty subset of the support has a least element. Addition is coefficientwise and multiplication is the convolution product. The well-ordering condition is exactly what makes every coefficient of a product a finite sum and keeps the support well ordered. The resulting field is denoted by

$$k((\Gamma)) = k((\Gamma)).$$

It is ordered lexicographically: the sign of a nonzero series is the sign of the coefficient at the least exponent in its support.

Theorem 1.1 (Hahn–Kaplansky). The ordered field $k((\Gamma))$ is real closed.

For $\Gamma = \mathbb{Q}$, define the bounded-denominator Puiseux field by

$$\mathcal{P}(k) = \left\{ x \in k((\mathbb{Q})) : \text{there is } N \geq 1 \text{ with } \operatorname{supp}(x) \subseteq \frac{1}{N}\mathbb{Z} \right\}.$$

The sum, product, and inverse of two elements with a common denominator bound have the same property, so $\mathcal{P}(k)$ is a subfield of $k((\mathbb{Q}))$.

Theorem 1.2 (Puiseux real closedness). The ordered field $\mathcal{P}(k)$ is real closed.

We use the standard characterization of real closed ordered fields; the valuation-theoretic background is standard, see [1, 2].

Proposition 1.3 (Ordered-field criterion). An ordered field K is real closed if the following two conditions hold:

(i) every $x \geq 0$ in K is a square;

(ii) every polynomial in $K[X]$ of odd degree has a root in K.

Thus both theorems reduce to the same two tasks. The square-root task is local at the least exponent of a series. The odd-degree root task is global: Newton corrections have to be summed, and their exponents have to be shown to remain under control.

2 Valuation preliminaries and square roots

For $x \neq 0$ in $k((\Gamma))$, put

$$v(x) = \min \operatorname{supp}(x), \qquad \operatorname{lc}(x) = a_{v(x)},$$

and put $v(0) = +\infty$. The least-exponent rule gives

$$v(xy) = v(x) + v(y), \qquad v(x+y) \geq \min\{v(x), v(y)\},$$

with equality in the second formula when the two valuations are different. The valuation ring and its maximal ideal are

$$\mathcal{O} = \{x : v(x) \geq 0\}, \qquad \mathfrak{m} = \{x : v(x) > 0\}.$$

The residue of an element of $\mathcal{O}$ is its coefficient at exponent zero. An element of $\mathcal{O}$ is a *unit* when its inverse also belongs to $\mathcal{O}$; equivalently, it has valuation zero. We call a series, or a polynomial coefficient, *integral* when it belongs to $\mathcal{O}$.

A family $(A_i)_{i\in I}$ of Hahn series is *Hahn summable* if $\bigcup_i \operatorname{supp}(A_i)$ is well ordered and, for every $\gamma \in \Gamma$, only finitely many i have $[t^\gamma]A_i \neq 0$. Its Hahn sum is then defined coefficientwise by those finite sums.

Lemma 2.1 (Leading-term normalization). If $x > 0$ in $k((\Gamma))$, then

$$x = c\,t^\gamma(1 + w)$$

for some $\gamma \in \Gamma$, $c \in k$ with $c > 0$, and $w \in \mathfrak{m}$.

Proof. Take $\gamma = v(x)$ and $c = \operatorname{lc}(x)$. The lexicographic order gives $c > 0$. Put $w = c^{-1}t^{-\gamma}x - 1$. The series $c^{-1}t^{-\gamma}x$ has valuation zero and leading coefficient one; hence its difference from 1 has positive valuation. Thus $w \in \mathfrak{m}$, and the stated identity follows from the definition of w. $\square$

An element u with $u - 1 \in \mathfrak{m}$, equivalently an element of the form $1 + w$ with $w \in \mathfrak{m}$, is called a 1-*unit*.

We use Neumann's support lemma: if S is a well-ordered subset of the positive cone of an ordered abelian group, then the additive monoid generated by S is well ordered, and a fixed group element has only finitely many representations as a sum of a finite sequence from S. Consequently the family of powers $(w^n)_{n\geq 0}$ is Hahn summable whenever $\operatorname{supp}(w) \subseteq \Gamma_{>0}$. This is the standard support lemma used in the construction of the Hahn field; see [2].

Lemma 2.2 (A 1-unit has a binomial square root). If $u \in k((\Gamma))$ satisfies $v(u - 1) > 0$, then there is $s \in k((\Gamma))$ with $u = s^2$.

Proof. Put $w = u-1$. If $w = 0$, take $s = 1$. Otherwise let $\epsilon = \min \operatorname{supp}(w) > 0$. The binomial series

$$s = \sum_{n\geq 0} \binom{1/2}{n} w^n$$

is a Hahn series. Indeed, Neumann's support lemma applies to the well-ordered positive set $\operatorname{supp}(w)$, so the union of the supports of the powers is well ordered and only finitely many powers contribute to any fixed exponent. Consequently the binomial identity can be checked coefficient by coefficient. It gives $s^2 = 1 + w = u$. $\square$

Proposition 2.3 (Nonnegative elements are squares). Every $x \geq 0$ in $k((\Gamma))$ is a square.

Proof. The assertion is clear for $x = 0$. For $x > 0$, write $x = c\,t^{\gamma}(1 + w)$ as in Lemma 2.1. Divisibility of Γ gives δ with $2\delta = \gamma$, and real closedness of k gives $d \in k^{\times}$ with $d^2 = c$. By Lemma 2.2, choose $s^2 = 1 + w$. Then

$$x = (d\,t^{\delta} s)^2.$$

$\square$

3 Newton reduction for odd-degree polynomials

Let $F(X) = \sum_i a_i X^i \in k((\Gamma))[X]$ and let $\delta \in \Gamma$. In the root construction $\delta > 0$ will be the exponent of a monomial correction ct^{δ}. The Newton weight of the i-th term is

$$W_{\delta}(i) = v(a_i) + i\delta.$$

For $\mu \in \Gamma$, write $[t^{\eta}]a$ for the coefficient of a Hahn series a at exponent η, and define the Newton polynomial on the weight level μ by

$$\mathrm{in}_{\delta,\mu}(F)(T) = \sum_{\substack{i:\ a_i \neq 0 \\ W_{\delta}(i) = \mu}} [t^{\mu - i\delta}] a_i\, T^i \in k[T].$$

Its nonzero terms are exactly the coefficient points on the line of weight μ. If $F \neq 0$ and $\mu_{\delta} = \min_{a_i \neq 0} W_{\delta}(i)$, then $\mathrm{in}_{\delta,\mu_{\delta}}(F)$ is the usual initial polynomial on the lowest Newton line; its coefficient in degree i is $\mathrm{lc}(a_i)$ whenever $W_{\delta}(i) = \mu_{\delta}$.

For a nonzero polynomial $P \in k[T]$, the *multiplicity* of a root c is the largest $r \in \mathbb{N}$ for which $(T - c)^r$ divides P. Equivalently, $P(c) = P'(c) = \cdots = P^{(r-1)}(c) = 0$ and $P^{(r)}(c) \neq 0$.

Lemma 3.1 (Normalization at an odd-multiplicity Newton root). Let $\mu \in \Gamma$, assume $\mu \le W_{\delta}(i)$ for every i, and suppose $c \in k$ is a root of $\mathrm{in}_{\delta,\mu}(F)$ of positive odd multiplicity r. Put

$$G(Z) = t^{-\mu} F\big(t^{\delta}(Z + c)\big) = \sum_j g_j Z^j.$$

Then

$$v(g_j) \ge 0, \qquad v(g_j) > 0\ (j < r), \qquad v(g_r) = 0.$$

Thus $G \in \mathcal{O}[Z]$, its degree-r coefficient is a unit, and all coefficients below degree r lie in $\mathfrak{m}$.

Proof. Write $F(ct^{\delta}+Y)=\sum_j b_j Y^j$. Expanding gives

$$F(ct^{\delta}+Y)=\sum_i a_i \sum_{j=0}^{i} \binom{i}{j} c^{i-j} t^{(i-j)\delta} Y^j.$$

The contribution to b_j at exponent $\mu - j\delta$ is $t^{\mu-j\delta}\,\mathrm{in}_{\delta,\mu}(F)^{(j)}(c)/j!$; after assigning weight δ to Y, the corresponding term has total weight μ. The multiplicity-r assumption makes these coefficients vanish for $j<r$ and makes the coefficient in degree r nonzero. All other contributions have strictly larger weighted value. Therefore

$$v(b_j)+j\delta \geq \mu, \qquad v(b_j)+j\delta > \mu \ (j<r), \qquad v(b_r)+r\delta=\mu.$$

Since $g_j = t^{j\delta-\mu} b_j$, the three weighted inequalities are exactly the three displayed valuation statements for G. Together with the assumed positivity and oddness of r, these are precisely the asserted coefficient conditions. $\square$

Definition 3.2 (Normalized odd Newton problem). A normalized odd Newton problem of multiplicity m consists of a polynomial $F \in k((\Gamma))[X]$, with $m>0$ odd, all of whose coefficients are integral, whose coefficient of X^m is a unit, and whose coefficients of degree below m lie in $\mathfrak{m}$. Equivalently, the coefficientwise lift of F to $\mathcal{O}[X]$ has these last two properties; we use that lift without changing notation whenever the valuation-ring formulation is needed.

Here a *translation* is a substitution $X \mapsto X+a$, and a *monomial dilation* is a substitution $X \mapsto t^{\delta}X$, possibly followed by multiplication of the polynomial by a nonzero monomial t^{η}. A solution of a normalized problem is an element $y \in \mathfrak{m}$ with $F(y)=0$. If the problem was obtained by finitely many such transformations, their inverses carry y to a root of the original polynomial.

The next lemma is the point at which real closedness of the residue field enters the Newton argument.

Lemma 3.3 (An odd initial polynomial has a nonzero odd-multiplicity root). Let $I \in k[T]$ have odd degree n and nonzero constant term. Then I has a nonzero root of odd multiplicity r, with $1 \leq r \leq n$.

Proof. Over a real closed field every irreducible factor has degree one or two. Since $\deg I$ is odd, some linear factor occurs with odd multiplicity; let its root be a and its multiplicity be r. The nonzero constant term says that 0 is not a root, hence $a \neq 0$. Positivity of root multiplicity and the elementary bound $r \leq \deg I = n$ give the remaining assertions. $\square$

For a normalized problem of multiplicity m with $v(a_0) = m\delta$, we call it the *lower-edge case at* δ when some $j < m$ satisfies $a_j \neq 0$ and $v(a_j) + j\delta < m\delta$. Thus a coefficient point lies below the line through the constant and degree-m points. A *proper multiplicity* means a multiplicity strictly smaller than m.

Proposition 3.4 (Lower Newton edges are resolved by multiplicity descent). Write $F(X) = \sum_i a_i X^i \in k((\Gamma))[X]$. Let m be odd (hence positive) and assume

$$v(a_m) = 0, \qquad v(a_i) > 0 \quad (0 \leq i < m), \qquad v(a_i) \geq 0 \quad (i > m).$$

Suppose every normalized odd Newton problem of odd multiplicity $r < m$ has a solution. Assume that for some $\delta \in \Gamma$

$$v(a_0) = m\delta$$

and that there is an index $j < m$ with $a_j \neq 0$ and

$$v(a_j) + j\delta < m\delta. \tag{LE}$$

Then F has a root of positive valuation.

Proof. For a slope s, call $w_s(i) = v(a_i) + is$ the Newton weight of the i-th coefficient. Thus (LE) says exactly that the point belonging to $a_j X^j$ lies strictly below the line of slope $-\delta$ through the constant and degree-m points: both endpoints have weight $m\delta$.

The lower point also proves that $\delta > 0$: rearranging its strict inequality gives $0 < v(a_j) < (m-j)\delta$, and $m - j > 0$.

Define the first lower slope by

$$\theta = \min_{\substack{0 \leq i < m \\ a_i \neq 0}} \frac{v(a_i)}{m - i}.$$

The set is nonempty by (LE). Every number in it is positive because the coefficients below degree m lie in $\mathfrak{m}$, while the index j in (LE) gives $\theta < \delta$. Hence

$$0 < \theta < \delta. \tag{S}$$

Consider

$$\widetilde{F}_\theta(T) = t^{-m\theta} F(t^\theta T) = \sum_i t^{(i-m)\theta} a_i T^i.$$

If $i < m$, the definition of θ gives $v(a_i) + (i - m)\theta \geq 0$. The same inequality is immediate for $i = m$, and for $i > m$ it follows from $a_i \in \mathcal{O}$ and $\theta > 0$. Thus $\widetilde{F}_\theta \in \mathcal{O}[T]$, and we may reduce it modulo $\mathfrak{m}$:

$$Q(T) = \overline{\widetilde{F}_\theta(T)} \in k[T].$$

The coefficient of T^m in Q is nonzero because $v(a_m) = 0$. All coefficients above degree m vanish after reduction because their valuations are strictly positive. Hence $\deg Q = m$. An index attaining the minimum defining θ gives a nonzero coefficient of Q below degree m. Finally,

$$v(a_0) - m\theta = m(\delta - \theta) > 0$$

by (S), so $Q(0) = 0$.

By the same linear–quadratic factorization argument as in Lemma 3.3, the odd-degree polynomial Q has a root a of positive odd multiplicity r; here we do not need that the root be nonzero. Necessarily $r < m$. Indeed, if $r = m$, then degree comparison gives $Q(T) = u(T - a)^m$ for some $u \neq 0$. Since $Q(0) = 0$, this forces $a = 0$, so $Q = uT^m$, contradicting the nonzero coefficient of degree $j < m$.

Now translate the root a to zero:

$$H(Z) = t^{-m\theta} F\big(t^\theta (Z + a)\big).$$

By Lemma 3.1, H is a normalized odd Newton problem of multiplicity $r < m$. The induction hypothesis gives $z \in \mathfrak{m}$ with $H(z) = 0$. Put

$$y = t^\theta (z + a).$$

Then $F(y) = t^{m\theta} H(z) = 0$. Moreover $z \in \mathfrak{m}$ and $a \in k$ imply $v(z + a) \geq 0$; together with $\theta > 0$, this gives $v(y) = \theta + v(z + a) > 0$. Thus y is the required root. □

For $m > 1$, the reduction of a normalized problem modulo $\mathfrak{m}$ has zero as a root of multiplicity exactly m: all coefficients below degree m vanish, while the degree-m coefficient does not. We call this the *same-multiplicity zero branch*. The phrase records this specific residue root and its multiplicity; it does not assert that the original polynomial already has a root.

4 Transfinite Newton continuation

We now solve the remaining same-multiplicity zero branch by a transfinite sequence of Newton corrections.

Definition 4.1 (Ordinal terminology). A well-order is a linear order in which every nonempty subset has a least element. An ordinal is the order type of a well-ordered set. If λ is an ordinal, its initial segment before $\alpha < \lambda$ consists of all $\beta < \alpha$. A successor ordinal has the form $\alpha + 1$; a nonzero ordinal with no greatest element is a limit ordinal. Transfinite recursion defines an object at a successor stage from the preceding object and at a limit stage from the whole earlier initial segment. We write Ord for the class of all ordinals.

At a Newton stage α, we retain an approximation $x_\alpha \in \mathfrak{m}$ and write

$$F_\alpha(Z) = F(x_\alpha + Z) = \sum_i b_i Z^i.$$

We say that this stage still has multiplicity m when F_α is a normalized odd Newton problem of multiplicity m in the sense of Definition 3.2. This is only terminology for the displayed translated polynomial, not an additional kind of Newton process.

Lemma 4.2 (Successor correction). Let $F \in \mathcal{O}[X]$, $m \in \mathbb{N}$, and $x \in \mathfrak{m}$. Write $F_x(Z) = F(x + Z) = \sum_i b_i Z^i$, and suppose $v(F(x)) = \gamma > 0$, $\rho > 0$, and $m\rho = \gamma$. Assume every Newton weight of F_x at ρ is at least γ, and let $c \in k^\times$ be a root of $\mathrm{in}_{\rho,\gamma}(F_x)$. Then

$$x' = x + ct^\rho \in \mathfrak{m}, \qquad v(x' - x) = \rho,$$

and $v(F(x')) > \gamma = v(F(x))$.

Proof. Both x and ct^ρ have positive valuation, so $x' \in \mathfrak{m}$, while $c \neq 0$ gives $v(x' - x) = \rho$. In the expansion of $F_x(ct^\rho)$, all terms have weight at least γ, and the coefficient at weight γ is precisely $\mathrm{in}_{\rho,\gamma}(F_x)(c) = 0$. Hence the entire weight-γ part cancels and $v(F(x')) > \gamma$. $\square$

At a normalized stage of multiplicity m, condition

$$v(b_i) + i\rho \geq \gamma \qquad (0 < i < m) \tag{NB}$$

together with integrality above degree m puts every weight on or above γ. The weight-level polynomial has odd degree m and nonzero constant term, so Lemma 3.3 supplies the nonzero root c required by Lemma 4.2.

For $m > 1$, a *fixed-lift, fixed-multiplicity Newton chain* is a natural-number sequence (x_n) in $\mathfrak{m}$, using one fixed integral coefficient lift of F, in which every successor is a correction from Lemma 4.2. In $x_{n+1} - x_n = c_n t^{\rho_n}$, we call ρ_n the *correction value* (or correction exponent).

Lemma 4.3 (Strict increase of successive correction values). In a fixed-lift, fixed-multiplicity Newton chain with $m > 1$, the map $n \mapsto \rho_n$ is strictly increasing.

Proof. The successor correction gives $v(F(x_{n+1})) > v(F(x_n))$. Writing these values as $m\rho_{n+1}$ and $m\rho_n$, respectively, and using that multiplication by the positive integer m is strictly order preserving gives the claim. $\square$

Lemma 4.4 (Hahn sum of a separated family). Let I be a well-ordered set. For every $i \in I$, let D_i be a Hahn series and let $\ell_i, u_i \in \Gamma$. Assume

$$\operatorname{supp}(D_i) \subseteq [\ell_i, u_i), \qquad i < j \Longrightarrow u_i \leq \ell_j. \tag{SS}$$

Then $(D_i)_{i \in I}$ is Hahn summable.

Proof. To see that $U = \bigcup_i \operatorname{supp}(D_i)$ is well ordered, take a nonempty subset $A \subseteq U$. Choose the least index i_0 for which $A \cap \operatorname{supp}(D_{i_0}) \neq \varnothing$, and then the least element a_0 of that intersection. No earlier support meets A, and (SS) puts every element of every later support at least $u_{i_0} > a_0$. Thus a_0 is the least element of A. Moreover a fixed exponent cannot belong to two distinct supports: if $i < j$, membership in both would give $\gamma < u_i \leq \ell_j \leq \gamma$. Hence only finitely many (in fact at most one) summands contribute to each coefficient, proving Hahn summability. $\square$

In particular, if λ is a limit ordinal and $(\rho_\alpha)_{\alpha<\lambda}$ is strictly increasing, apply the lemma with $D_\alpha = c_\alpha t^{\rho_\alpha}$, $\ell_\alpha = \rho_\alpha$, and $u_\alpha = \rho_{\alpha+1}$. Since $\alpha+1 < \lambda$, this proves that $x_0 + \sum_{\alpha<\lambda} c_\alpha t^{\rho_\alpha}$ is a Hahn series.

For a Hahn-summable family $A = (A_*) \cup (A_i)_{i \in I}$, a polynomial $G(T) = \sum_{j=0}^{d} b_j T^j$, and $\gamma \in \Gamma$, define $E_{G,\gamma}(A)$ as follows. Expand each coefficient $[t^\gamma] b_j (\sum A)^j$; collect every ordinary index $i \in I$ that occurs in a tuple with nonzero contribution, and take the union over $0 \leq j \leq d$. Hahn summability makes this a finite set. The distinguished index $*$ is not included in $E_{G,\gamma}(A)$.

Lemma 4.5 (Finite dependence of polynomial-evaluation coefficients). Let $A = (A_*) \cup (A_i)_{i \in I}$ be a Hahn-summable family, where the distinguished term A_* is always retained. For a predicate p on I, write $A|_p$ for the family obtained by replacing A_i by zero when $p(i)$ is false. Given $G \in k((\Gamma))[T]$ and $\gamma \in \Gamma$, one has

$$E_{G,\gamma}(A) \subseteq \{i : p(i)\} \quad \Longrightarrow \quad [t^\gamma] G\Big(\sum A\Big) = [t^\gamma] G\Big(\sum (A|_p)\Big). \tag{FD}$$

Proof. By the preceding definition, if p contains $E_{G,\gamma}(A)$, truncating away indices outside p removes none of the contributing tuples, so every summand in the coefficient expansion is unchanged. This proves (FD). For an ordinal correction family, any initial segment containing this finite set is therefore sufficiently late. $\square$

On the multiplicity-1 side, a whole ω-block of natural-number corrections, rather than one monomial correction, is used as one stage of the transfinite recursion. Fix $F \in \mathcal{O}[X]$. The block data below make no coefficient or multiplicity assumption on F; those assumptions enter only when the data are constructed. For a nonroot $x \in \mathfrak{m}$, meaning $F(x) \neq 0$, write $v(x) = v(F(x)) \in \Gamma$.

A *Newton block successor* is a move from a nonroot $x \in \mathfrak{m}$ to a nonroot $x^+ \in \mathfrak{m}$ such that

$$v(x) < v(x^+), \qquad \operatorname{supp}(x^+ - x) \subseteq [v(x), v(x^+)).$$

It is the compressed datum of the limit of one bounded natural correction chain. A further correction from that limit is used only to certify that the limit is a rootless state from which the construction can continue; that further correction is not part of the block difference.

Lemma 4.6 (A root or a multiplicity-1 block successor). Let $F \in \mathcal{O}[X]$, assume its constant coefficient lies in $\mathfrak{m}$ and its linear coefficient is a unit, and let $x \in \mathfrak{m}$ be a nonroot. Then either F has a root in $\mathfrak{m}$, or there is a Newton block successor from x.

Proof. Translate by x. Its constant coefficient is $F(x) \in \mathfrak{m}$, and its linear coefficient is congruent modulo $\mathfrak{m}$ to the original unit coefficient; all contributions from higher degrees contain a positive-valuation power of x. Thus the translated polynomial is again a normalized multiplicity-1 problem. Unless a root occurs, repeatedly apply Lemma 4.2. This gives corrections $c_n t^{\rho_n}$ with strictly increasing positive values and finite approximations $x_n = x + \sum_{j<n} c_j t^{\rho_j}$, where $v(F(x_n)) = \rho_n$.

The correction monomials are Hahn summable; put $x_\omega = x + \sum_n c_n t^{\rho_n}$. The tail $x_\omega - x_{n+1}$ has valuation ρ_{n+1}. Taylor expansion, with integral coefficients and integral arguments, gives

$$v\big(F(x_\omega) - F(x_{n+1})\big) \geq \rho_{n+1}.$$

Since $v(F(x_{n+1})) = \rho_{n+1}$, it follows that $v(F(x_\omega)) \geq \rho_{n+1} > \rho_n$. If the ρ_n are unbounded, this says every coefficient of $F(x_\omega)$ is zero, so x_ω is the first alternative. Otherwise, if no root has occurred, write the finite positive value of $F(x_\omega)$ as η. It satisfies $\rho_n < \eta$ for every n.

At multiplicity 1, the non-below condition is empty. For the translated polynomial at x_ω, take correction value $\delta = \eta$. Its weight-level polynomial has nonzero constant and linear terms and hence a nonzero root. One further application of Lemma 4.2 therefore gives a correction from x_ω. In the Lean construction this extra step certifies the bounded omega-block, but its target is not the block endpoint.

Set $x^+ = x_\omega$. It is a nonroot in $\mathfrak{m}$, $v(x^+) = \eta > \rho_0 = v(x)$, and the support of $x^+ - x$ consists of the ρ_n's. Since $\rho_0 \leq \rho_n < \eta$,

$$\operatorname{supp}(x^+ - x) \subseteq [v(F(x)), v(F(x^+))).$$

Thus $x \mapsto x^+$ is the required Newton block successor. $\square$

Definition 4.7 (Coherent multiplicity-1 Newton block process). Let I be a well-ordered set with least element 0. A *coherent multiplicity-1 Newton block process* on I consists, for every $\alpha \in I$, of a nonroot $x_\alpha \in \mathfrak{m}$ and a Newton block successor x_α^+, such that $\alpha < \beta$ implies

$$v(x_\alpha^+) \leq v(x_\beta),$$

and, for every α,

$$x_\alpha - x_0 = \sum_{\xi < \alpha} (x_\xi^+ - x_\xi). \tag{BC}$$

The first condition separates the supports of distinct blocks; the second says that every state is the Hahn sum of all preceding block corrections.

We shall use the following compatibility package at a nonzero limit ordinal λ. For every $\tau < \lambda$, suppose a coherent block process $C^{<\tau}$ has been constructed on the stages below τ. From $C^{<\alpha+1}$, select its final state and final block and call them x_α and B_α, and write x_α^+ for the endpoint of B_α. The family has *natural limit compatibility* when the following conditions hold.

(i) There is one $x_0 \in \mathfrak{m}$ which is the bottom source of every nonempty $C^{<\tau}$, and the selected state at index 0 also has source x_0.

(ii) If $\alpha < \beta < \lambda$, the value at the endpoint of B_α equals the endpoint value of the corresponding block after $C^{<\beta+1}$ is restricted through α.

(iii) If $\alpha < \beta < \lambda$, the Hahn sum of the selected blocks before α equals the Hahn sum of the restriction of $C^{<\beta+1}$ before α.

(iv) The latter restricted sum equals the prefix sum formed inside $C^{<\alpha+1}$ itself.

Conditions (iii) and (iv) are genuine compatibility assumptions; support separation alone does not imply that independently constructed prefixes have the same coefficients.

On the higher-multiplicity side, the transfinite state retains the bounded natural correction chain itself. For a normalized problem F of multiplicity m, an *integral coefficient lift* is a polynomial $D \in \mathcal{O}[X]$ which equals F in the Hahn field, whose coefficients below m lie in $\mathfrak{m}$, and whose degree-m coefficient is a unit. With one such D fixed, a *Newton correction step* from $A \in \mathfrak{m}$ to $A' \in \mathfrak{m}$ consists of $\gamma, \delta \in \Gamma$ and $c \in k$ such that

$$\begin{aligned} &\gamma > 0, \quad \delta > 0, \quad c \neq 0, \quad m\delta = \gamma, \quad A' = A + ct^{\delta}, \\ &v(D(A)) = \gamma, \qquad v(A' - A) = \delta, \qquad v(D(A')) > \gamma. \end{aligned} \tag{NS}$$

The *root-or-step property* says the following for every normalized same-multiplicity zero branch (F, m), after roots at all multiplicities below m have been supplied: there is an integral coefficient lift D such that, from every $A \in \mathfrak{m}$, either F already has a root in $\mathfrak{m}$, or there is a Newton correction step from A, using that same D. Its *fixed-lift form* requires this alternative for every integral coefficient lift D. The two forms agree here: two integral lifts have the same image in the Hahn field, and the inclusion $\mathcal{O} \hookrightarrow k((\Gamma))$ is injective, so the lifts are equal.

A sequence $(x_n)_{n\in\mathbb{N}}$ is a *rootless natural correction chain* if $x_0 = 0$, all x_n lie in $\mathfrak{m}$, and one integral coefficient lift D makes every move $x_n \to x_{n+1}$ a Newton correction step. It is *bounded* when its correction values are bounded above in Γ. A *terminal obstruction* is a same-multiplicity zero branch equipped with such a bounded chain and its fixed lift D.

A *shifted terminal state* over the original polynomial F consists of a shift $s \in \mathfrak{m}$, the translated polynomial $F_s(Z) = F(s + Z)$, a terminal obstruction for F_s, and the transport taking a root z of F_s to the root $s + z$ of F. The first correction value of the natural chain in this state is called its *rank*.

Definition 4.8 (Coherent higher-multiplicity terminal-state process). Let I be a linearly ordered set, let $F \in k((\Gamma))[X]$, and let $m \in \mathbb{N}$. Write $(S_\alpha)_{\alpha\in I}$ for shifted terminal states for (F, m), s_α for their shifts, and ρ_α for their ranks. (The terminal data in each state already imply that $m > 1$ is odd.) The family is *coherent* when $\alpha < \beta$ implies $\rho_\alpha < \rho_\beta$, and whenever $\alpha \leq \beta$ and $\gamma < \rho_\alpha$,

$$[t^\gamma]s_\alpha = [t^\gamma]s_\beta. \tag{TC}$$

Stability of polynomial-evaluation coefficients is not included as a field of this definition; it is derived as the recursion invariant below.

Proposition 4.9 (The multiplicity-1 limit prefix). Let λ be a nonzero limit ordinal. Suppose that a process from Definition 4.7 has been constructed on every proper initial segment and that the resulting family has natural limit compatibility. Then the selected states x_α and blocks B_α, for $\alpha < \lambda$, form a coherent Newton block process on the whole initial segment below λ.

Proof. By (ii), the endpoint value of an earlier selected block is the endpoint value of its copy in every later process. The separation condition in the later process therefore gives $v(x_\alpha^+) \le v(x_\beta)$ whenever $\alpha < \beta$. Hence the selected blocks have separated supports and are Hahn summable by Lemma 4.4.

It remains to check source coherence. Fix $\alpha < \lambda$ and choose β with $\alpha < \beta < \lambda$, which is possible because λ is a limit ordinal. Coherence inside $C^{<\beta+1}$ identifies the difference between its state at α and its bottom source with its restricted prefix sum. By (i) the bottom source is x_0, by (iii) the restricted sum is the selected-block sum before α, and by (iv) this is also the prefix already computed in $C^{<\alpha+1}$. Thus

$$x_\alpha - x_0 = \sum_{\xi<\alpha} (x_\xi^+ - x_\xi),$$

which is exactly (BC). The state, block-successor, and nonroot fields are inherited from the selected final data. This constructs the required coherent process. □

Proposition 4.10 (Invariant of the higher-multiplicity terminal process). Fix a terminal obstruction for (F, m); its branch data imply that $m > 1$ is odd. Assume that every normalized problem of multiplicity below m has been solved, that the fixed-lift root-or-step property holds, and that F has no root in $\mathfrak{m}$. Let $(S_\beta)_{\beta\in\mathrm{Ord}}$ be the recursively defined family, with shifts s_β and ranks ρ_β. Then for every ordinal o:

(i) $\beta < o$ implies $\rho_\beta < \rho_o$;

(ii) if $\beta \le o$ and $\gamma < \rho_\beta$, then $[t^\gamma]s_o = [t^\gamma]s_\beta$;

(iii) if $\beta \le o$ and $\gamma < m\rho_\beta$, then

$$[t^\gamma]F(s_o) = [t^\gamma]F(s_\beta); \tag{EC}$$

(iv) the map $\beta \mapsto \rho_\beta$ on $\beta < o$ is strictly increasing and

$$s_o = \sum_{\xi < o} (s_{\xi+1} - s_\xi). \tag{HS}$$

In particular, restriction to any ordinal initial segment is coherent in the sense of Definition 4.8.

Proof. Use transfinite induction on β. At 0, take the original terminal obstruction with shift $s_0 = 0$; all four assertions are immediate.

Suppose S_α has been constructed. A lower edge at its first correction value would give a root by Proposition 3.4 and the solved lower-multiplicity problems; translating back would contradict the no-root assumption. Thus there is no lower edge. Advance by the first correction of the bounded chain and replace the chain by its tail, expressed relative to the new source. The shift increment is a nonzero monomial of value ρ_α, the new chain is bounded, and its first correction value is strictly larger than ρ_α. The Newton weight estimate behind (NS) also says that this move preserves the coefficients of $F(s_\alpha)$ below $m\rho_\alpha$. These facts give rank growth, shift-coefficient stability, evaluation-coefficient stability, and the prefix-sum identity at $\alpha + 1$.

Now let β be a nonzero limit ordinal. By induction the increments $s_{\xi+1} - s_\xi$ are nonzero monomials of the strictly increasing values ρ_ξ. They are Hahn summable by Lemma 4.4; define s_β to be their sum. This proves (HS), places s_β in $\mathfrak{m}$, and gives (TC): below ρ_α, every increment from stage α onward has zero coefficient. For $\alpha < \beta$ and $\gamma < m\rho_\alpha$, the finite set from Lemma 4.5 is contained in some initial segment $\eta < \beta$. Taking $\eta \geq \alpha$ and using the induction hypothesis at η proves (EC) at the limit.

The translated polynomial is still a same-multiplicity zero branch. Indeed, translation by $s_\beta \in \mathfrak{m}$ sends every coefficient below degree m into $\mathfrak{m}$: in the Taylor expansion, the old coefficients below m already lie in $\mathfrak{m}$, while every term coming from degree at least m contains a positive power of s_β. Its degree-m coefficient is the old unit plus terms in $\mathfrak{m}$, and hence remains a unit. Oddness and $m > 1$ are unchanged. Thus the hypotheses needed at the limit state have not been lost.

Apply the fixed-lift root-or-step property at the translated polynomial $F(X + s_\beta)$. Under the no-root assumption it supplies a natural correction chain. If its values σ_n were unbounded, its monomial corrections would have a Hahn sum z. Given $\gamma \in \Gamma$, take a stage after all indices in the finite-dependence set for the coefficient $[t^\gamma]F(s_\beta + z)$, and then take it still later so that $m\sigma_n > \gamma$. The coefficient equals that at this finite approximation, whose evaluation has valuation $m\sigma_n > \gamma$, so it is zero. Thus every coefficient vanishes and $F(s_\beta + z) = 0$, a contradiction. The new chain is therefore bounded and supplies the terminal state at β.

It remains to prove strict rank growth into the limit. Given $\alpha < \beta$, choose $\alpha < \eta < \beta$. The evaluation-coefficient stability just proved says that every coefficient of $F(s_\beta)$ below $m\rho_\eta = v(F(s_\eta))$ is zero. Therefore $m\rho_\eta \le v(F(s_\beta)) = m\rho_\beta$. Hence $\rho_\alpha < \rho_\eta \le \rho_\beta$. This completes all induction clauses. ☐

Definition 4.11 (Hartogs ordinal). For a set X, let $\#X$ denote its cardinality and let $(\#X)^+$ be the successor cardinal, the least cardinal strictly larger than $\#X$. For a cardinal κ, let $\operatorname{ord}(\kappa)$ be the least ordinal of cardinality κ. Define the *Hartogs ordinal* by

$$h(X) = \operatorname{ord}\big((\#X)^+\big).$$

We write $|h(X)|$ for its underlying well-ordered set of predecessors.

The underlying set $|h(X)|$ has cardinality $(\#X)^+$. An injection $|h(X)| \to X$ would therefore imply $(\#X)^+ \le \#X$, contradicting the defining inequality $\#X < (\#X)^+$. Thus this definition has the usual Hartogs property: it produces a well-ordered set which cannot inject into X. Equivalently, it is the least initial ordinal with that property. This cardinal argument is the only set-theoretic fact used below.

Lemma 4.12 (Hartogs obstruction). There is no strictly increasing map from $|h(\Gamma)|$ into the ordered set Γ.

Proof. A strictly increasing map is injective, while the cardinal argument after Definition 4.11 says that no injection from $|h(\Gamma)|$ into the underlying set of Γ exists. ☐

Lemma 4.13 (Resolution of a terminal obstruction). Fix a terminal obstruction for (F, m), assume every normalized odd problem of multiplicity below m has been solved, and assume the root-or-step property defined above. Then the obstruction is resolved: F has a root in $\mathfrak{m}$.

Proof. Suppose, for a contradiction, that the original problem has no root in $\mathfrak{m}$. A root of any translated problem would transport to a root of the original one, so the root alternative is excluded at every stage. The root-or-step property initially selects some integral lift. For any prescribed lift, both lifts map to the same Hahn-field polynomial; injectivity of $\mathcal{O} \hookrightarrow k((\Gamma))$ makes them equal. Hence the property has the fixed-lift form required by

Proposition 4.10. Applying that proposition to the initial terminal obstruction constructs a shifted terminal state S_α for every $\alpha \in |h(\Gamma)|$, with strictly increasing ranks ρ_α. These ranks define a strictly increasing map

$$|h(\Gamma)| \longrightarrow \Gamma,$$

contradicting Lemma 4.12. The assumption that no root exists is therefore false. $\square$

Proposition 4.14 (The multiplicity-one case). A normalized odd Newton problem of multiplicity one has a solution in $\mathfrak{m}$.

Proof. Assume that no root exists. Zero is then a nonroot state. At every successor ordinal, Lemma 4.6 supplies the next block. At a limit ordinal, the already constructed objects are restrictions of this one recursion: their bottom source, corresponding endpoint values, and prefix Hahn sums therefore satisfy natural limit compatibility. Apply Proposition 4.9 to obtain the coherent process on the whole limit initial segment. Its separated block differences are Hahn summable, so define the limit source by

$$x_\lambda = x_0 + \sum_{\xi<\lambda}(x_\xi^+ - x_\xi).$$

Every block difference has positive support, hence $x_\lambda \in \mathfrak{m}$. The separation condition also extends to this new source. For $\alpha < \lambda$, put $\beta = \alpha + 1 < \lambda$. Coherence gives $x_\beta = x_\alpha^+$, and every block in the tail $x_\lambda - x_\beta$ has support at least $v(x_\beta)$. Thus $v(x_\lambda - x_\beta) \geq v(x_\beta)$. Taylor expansion over $\mathcal{O}$ gives

$$v(F(x_\lambda) - F(x_\beta)) \geq v(x_\beta), \qquad v(F(x_\lambda)) \geq v(x_\beta) = v(x_\alpha^+).$$

Under the standing no-root assumption x_λ is a nonroot, so this is the required inequality $v(x_\alpha^+) \leq v(x_\lambda)$. Applying Lemma 4.6 at x_λ supplies the top block for the next successor initial segment. Transfinite recursion therefore continues for as long as no root appears.

The separation inequality in Definition 4.7, together with strict increase inside every block, makes the residual values $\alpha \mapsto v(x_\alpha)$ strictly increasing. If the construction continued through all of $|h(\Gamma)|$, these values would define a strictly increasing map from that ordinal into Γ, contradicting Lemma 4.12. A root must therefore occur, and all corrections have positive value, so that root lies in $\mathfrak{m}$. $\square$

Proposition 4.15 (Termination in the higher-multiplicity branch). Let $m > 1$ be odd and assume that every normalized problem of proper odd multiplicity below m has a solution. Assume also the root-or-step property stated above. Then a normalized problem whose residue root at zero has multiplicity m has a solution in $\mathfrak{m}$.

Proof. Suppose there were no solution. Repeated application of the root-or-step property gives one integral lift and a rootless natural correction chain (x_n), with correction values ρ_n: a lower edge would be solved by Proposition 3.4 and the proper-multiplicity hypothesis, while otherwise Lemma 4.2 supplies the next term. Suppose first that (ρ_n) is unbounded. Its strictly increasing correction monomials are Hahn summable; write their sum as x. Fix $\gamma \in \Gamma$. First go beyond every index in the finite set from Lemma 4.5, and then go still farther, if necessary, to choose n with $m\rho_n > \gamma$. The coefficient in $F(x)$ at t^γ is then the same as that of this finite approximation. But

$$v(F(x_n)) = m\rho_n > \gamma,$$

so that coefficient is zero. Since γ was arbitrary, $F(x) = 0$, contrary to the assumption.

Thus the correction values are bounded and the chain, together with the fixed coefficient data that produces its corrections, is a terminal obstruction. By Lemma 4.13, that obstruction itself yields a solution in $\mathfrak{m}$, a contradiction. □

5 Odd roots and real closedness of the Hahn field

Lemma 5.1 (A slope dominated by the top-degree term). Assume that Γ is nontrivial. Let $0 \neq F(X) = \sum_i a_i X^i \in k((\Gamma))[X]$, put $n = \deg F$, and write $\gamma_n = v(a_n)$. There is $\delta \in \Gamma$ such that, for every $i < n$ with $a_i \neq 0$,

$$(n-i)\delta < v(a_i) - \gamma_n. \tag{DS}$$

Proof. There are only finitely many indices $i < n$ with $a_i \neq 0$. For each of them, divisibility of Γ gives a unique threshold q_i with

$$(n-i)q_i = v(a_i) - \gamma_n.$$

Uniqueness follows because an ordered abelian group is torsion-free. Choose $\varepsilon > 0$, which exists because Γ is nontrivial. If the set of thresholds is nonempty, take $\delta = \min_i q_i - \varepsilon$; if it is empty, take any δ. Then $\delta < q_i$ is exactly (DS). □

Theorem 5.2 (Odd-degree roots in the Hahn field). Every polynomial in $k((\Gamma))[X]$ of odd degree has a root in $k((\Gamma))$.

Proof. First prove by strong induction on odd $m > 0$ that every normalized problem of multiplicity m has a root in $\mathfrak{m}$. The case $m = 1$ is Proposition 4.14. If $m > 1$, the induction hypothesis solves every proper odd multiplicity below m. Translate at an arbitrary current approximation in $\mathfrak{m}$. If a coefficient lies below the current Newton edge, Proposition 3.4 gives a root. Otherwise the weight inequalities in (NB), Lemma 3.3, and Lemma 4.2 give the next Newton correction. Fixing the integral coefficient lift therefore gives the root-or-step property required by Proposition 4.15, which supplies a root. This completes the induction.

Now let F have odd degree n. If Γ is trivial, every Hahn series has support at the sole group element, so the Hahn field identifies with the real closed coefficient field k; the required root then exists in k. Suppose Γ is nontrivial. Since n is odd, $F \neq 0$ and $n > 0$. Write $F = \sum_i a_i X^i$ and $\gamma_n = v(a_n)$. Apply Lemma 5.1 to obtain δ, put $\lambda = \gamma_n + n\delta$, and define $\widetilde{F}(X) = t^{-\lambda} F(t^\delta X)$. Its degree-$n$ coefficient has value $-\lambda + \gamma_n + n\delta = 0$, while for $i < n$ with $a_i \neq 0$ the degree-i coefficient has value

$$-\lambda + v(a_i) + i\delta = v(a_i) - \gamma_n - (n - i)\delta > 0.$$

Coefficients above degree n vanish. Thus $\widetilde{F}$ is a normalized problem of odd multiplicity n. The induction just proved gives $z \in \mathfrak{m}$ with $\widetilde{F}(z) = 0$. Therefore

$$F(t^\delta z) = t^\lambda \widetilde{F}(z) = 0,$$

so $t^\delta z$ is a root of the original polynomial. $\square$

Proof of Theorem 1.1. Proposition 2.3 gives square roots of all nonnegative elements, and Theorem 5.2 gives roots of all odd-degree polynomials. Proposition 1.3 now applies. $\square$

6 Newton–Puiseux construction with bounded denominators

For $N \geq 1$, put

$$\Gamma_N = \frac{1}{N}\mathbb{Z}, \qquad \mathcal{P}_N(k) = \{x \in k((\mathbb{Q})) : \operatorname{supp}(x) \subseteq \Gamma_N\}.$$

Then $\mathcal{P}(k) = \bigcup_{N \geq 1} \mathcal{P}_N(k)$. A finite collection of Puiseux series and polynomial coefficients is contained in one common $\mathcal{P}_N(k)$.

The argument is the classical Newton–Puiseux construction. We record the extra denominator control needed for roots in the union of the rings $\mathcal{P}_N(k)$; see [3].

We call $\mathcal{P}_N(k)$ the *fixed denominator level* N. For a polynomial over its valuation ring, the *residue polynomial* is obtained by reducing every coefficient modulo the maximal ideal. A residue root $a \in k$ is *simple* when the residue polynomial vanishes at a but its derivative does not. In a Newton step of current multiplicity M, the *full-multiplicity case* is the case in which the chosen root of the initial polynomial has multiplicity exactly M; a smaller multiplicity is proper.

For a local ring with maximal ideal $\mathfrak{n}$, its *maximal-ideal topology* is the topology in which congruence modulo $\mathfrak{n}^r$, for $r = 1, 2, \ldots$, gives a neighborhood basis of zero.

Definition 6.1 (Power-series model of a fixed denominator level). For $N \geq 1$, put $q = t^{1/N}$ and

$$\mathcal{O}_N = \mathcal{P}_N(k) \cap \mathcal{O}$$

. The *power-series identification* of this level is the coefficient re-indexing isomorphism

$$\Phi_N : k[[q]] \xrightarrow{\sim} \mathcal{O}_N, \qquad \sum_{i \geq 0} a_i q^i \longmapsto \sum_{i \geq 0} a_i t^{i/N}.$$

Indeed, the image support is well ordered. Conversely, every exponent in an element of $\mathcal{O}_N$ is nonnegative and belongs to $\frac{1}{N}\mathbb{Z}$, hence is uniquely i/N for some $i \in \mathbb{N}$. Re-indexing by this unique i is the inverse map; addition and convolution multiplication are preserved because $(i + j)/N = i/N + j/N$.

Lemma 6.2 (Completeness of a fixed denominator level). The ring $\mathcal{O}_N$ is complete for its maximal-ideal topology.

Proof. Under the power-series identification from Definition 6.1, the maximal ideal of $\mathcal{O}_N$ corresponds to (q). Thus congruence modulo $(q)^r$ means equality of the coefficients of $1, q, \ldots, q^{r-1}$. A Cauchy sequence therefore has each coefficient eventually constant. Taking those eventual values produces a series in $k[[q]]$, and the sequence converges to it modulo every $(q)^r$. Transporting this limit through Φ_N proves the asserted completeness of $\mathcal{O}_N$. $\square$

Lemma 6.3 (Simple roots lift without changing the denominator). Let $F \in \mathcal{O}_N[X]$, and suppose that its residue polynomial has a simple root $a \in k$. Then F has a root in $\mathcal{O}_N$ reducing to a.

Proof. This is the simple-root form of Hensel's lemma for a complete local ring. We include its Newton argument. Start with a lift x_0 of a. Then $F(x_0) \in (q)$, while $F'(x_0) \notin (q)$, so $F'(x_0)$ is a unit. Define Newton's iteration

$$x_{n+1} = x_n - \frac{F(x_n)}{F'(x_n)}$$

and put $e_n = F(x_n)$. Taylor expansion in the polynomial ring gives

$$F(x_n + h) = F(x_n) + F'(x_n)h + h^2 R_n(h)$$

for a polynomial R_n over $\mathcal{O}_N$. With $h = -e_n/F'(x_n)$, the first two terms cancel, hence

$$v(e_{n+1}) \geq 2v(e_n), \qquad v(x_{n+1} - x_n) = v(e_n).$$

Moreover $x_{n+1} \equiv x_n \pmod{(q)}$, so $F'(x_{n+1}) \equiv F'(x_n) \not\equiv 0 \pmod{(q)}$; every derivative used above remains a unit. Since $v(e_0) \geq 1/N$, induction yields $v(e_n) \geq 2^n/N$, and therefore (x_n) is Cauchy for the (q)-adic topology. By Lemma 6.2 it converges to some $x \in \mathcal{O}_N$. The inequalities $v(e_n) \to \infty$ and continuity of polynomial evaluation give $F(x) = 0$, while $x \equiv a \pmod{(q)}$. $\square$

Lemma 6.4 (Full multiplicity preserves the denominator lattice). Let $F \in \mathcal{O}[X]$ and $y \in \mathcal{O}$, and suppose that the supports of every coefficient of F and of y lie in Γ_N. Let $M > 0$, $\delta > 0$, and $\gamma = M\delta$. Assume that the coefficient of X^M in $F(X+y)$ is a unit and that $\mathrm{in}_{\delta,\gamma}(F(X+y))$ has a nonzero root $c \in k^\times$ of multiplicity exactly M. Then $\delta \in \Gamma_N$.

Proof. For $i > M$, the translated coefficient is integral and hence its Newton weight is at least $i\delta > M\delta = \gamma$. Thus the initial polynomial has degree at most M, while its degree-M coefficient is nonzero because that translated coefficient is a unit. Its degree is therefore exactly M. A root of multiplicity M in a degree-M polynomial forces the factorization

$$I(T) = u(T-c)^M, \qquad u \neq 0.$$

Its coefficient of T^{M-1} is $-uMc \neq 0$. Since the translated degree-M coefficient is a unit, its exponent on the weight level γ is $0 = \gamma - M\delta$. The coefficient of T^{M-1} comes from exponent $\gamma - (M-1)\delta$. This exponent is therefore equal to δ. It belongs to the support of the degree-$(M-1)$ coefficient of $F(X+y)$. Translation by y preserves Γ_N-supported coefficients, because only finite sums and products occur. Hence this translated coefficient, and therefore δ, lies in Γ_N. $\square$

Lemma 6.5 (A strict chain in a rational lattice is unbounded). Let $f : \mathbb{N} \to \mathbb{Q}$. If f is strictly increasing and $f(n) \in \Gamma_N$ for every n, then f is not bounded above.

Proof. More precisely,

$$f(n+1) - f(n) \geq \frac{1}{N}, \qquad f(n) \geq f(0) + \frac{n}{N}.$$

The difference of two elements of Γ_N is again in Γ_N, and every positive element of Γ_N is at least $1/N$. Summing gives the second inequality. If b were an upper bound, choose $n > N(b - f(0))$; then $f(0) + n/N > b$, a contradiction. □

Theorem 6.6 (Bounded-denominator odd roots). Every odd-degree polynomial over $\mathcal{P}(k)$ has a root in $\mathcal{P}(k)$.

Proof. Let P be the given polynomial. Put all its coefficients in one denominator level. Write $d = \deg P$ and let γ_d be the value of its leading coefficient. Apply Lemma 5.1 with $\Gamma = \mathbb{Q}$, choose δ, and put $\lambda = \gamma_d + d\delta$. Enlarge the denominator once so that $\delta, \lambda \in \Gamma_N$. The same coefficient calculation as in the proof of Theorem 5.2 shows that

$$F(X) = t^{-\lambda} P(t^{\delta} X)$$

is a normalized problem all of whose coefficients lie in $\mathcal{P}_N(k)$. We solve F by strong induction on its odd multiplicity. A simple residue root is lifted by Lemma 6.3. A proper odd multiplicity $r < M$ is normalized by one translation and one monomial dilation. Only finitely many new rational monomial exponents occur. The scalar constants lie in k and have support at zero, while the existing coefficients already have a common denominator. Thus, after replacing N by one common multiple of the old and new denominators, all transformed coefficients lie in a single denominator level. The induction hypothesis at r gives a root at some bounded level, and undoing these two finite transformations still has bounded denominator. Thus denominator enlargement in every multiplicity-decreasing branch is finite; an infinite branch can only be the full-multiplicity branch treated next.

In the full-multiplicity case of current multiplicity M, Lemma 6.4 shows inductively that the translated polynomial and every successive correction remain at the same fixed level Γ_N. If the corrections terminate, the finite sum of corrections is a root in $\mathcal{P}_N(k)$. Otherwise write the corrections as $c_n t^{\delta_n}$. The values δ_n are strictly increasing in Γ_N, hence are unbounded by Lemma 6.5.

The family $(c_n t^{\delta_n})_n$ is Hahn summable and its sum x still has support in Γ_N. Fix an exponent γ. First go beyond every index in the finite set supplied by Lemma 4.5, and then increase n, if necessary, until $\delta_n > \gamma$. The coefficient of t^γ in $F(x)$ is then the coefficient at the n-th approximation. At that stage the successor construction gives $v(F(x_n)) = M\delta_n$. Since $M > 0$ and $\delta_n > \gamma$, this valuation is greater than γ, so the coefficient of t^γ in $F(x_n)$, and hence in $F(x)$, is zero. Thus every coefficient of $F(x)$ is zero and $F(x) = 0$. Consequently $x \in \mathcal{P}_N(k)$, and $t^\delta x \in \mathcal{P}(k)$ is a root of the original polynomial P. □

Proposition 6.7 (Nonnegative Puiseux series are squares). Every nonnegative element of $\mathcal{P}(k)$ is a square in $\mathcal{P}(k)$.

Proof. The case $x = 0$ is immediate. If $0 < x \in \mathcal{P}_N(k)$, the leading-term argument of Section 2 halves the leading exponent and the binomial series uses only sums of exponents in Γ_N. Thus the constructed square root has support in Γ_{2N}, and hence belongs to $\mathcal{P}(k)$. □

Proof of Theorem 1.2. Apply Proposition 6.7, Theorem 6.6, and Proposition 1.3. □

7 Corollaries with real coefficients

We use the classical fact that $\mathbb{R}$ is real closed. Equivalently, every nonnegative real number has a real square root and every odd-degree real polynomial has a real zero.

Corollary 7.1 (The real rational Hahn field). The ordered Hahn field $\mathbb{R}((\mathbb{Q}))$ is real closed.

Proof. The ordered additive group $\mathbb{Q}$ is divisible: for $q \in \mathbb{Q}$ and $n > 0$, the element q/n satisfies $n(q/n) = q$. Theorem 1.1, with $k = \mathbb{R}$ and $\Gamma = \mathbb{Q}$, therefore applies. □

Corollary 7.2 (Real Puiseux field). The field $\mathcal{P}(\mathbb{R})$ is real closed.

Proof. Apply Theorem 1.2 with $k = \mathbb{R}$. □

A Correspondence with the Lean formalization

The main text is deliberately mathematical and uses no Lean terminology. This appendix records the correspondence needed to audit that its case splits, induction parameter, limit construction, and denominator control agree with the maintained formal proof. Declaration and module names here are audit markers, not substitutes for arguments in the main text. Each assertion in the left column links back to its complete statement and proof. `HahnField.exists_leadingTerm_decomposition_of_pos`, `KSameMultiplicityOrdinalJetChain`, `iterateFixedLiftTerminalOrdinalPrefixState_invariant`, and `not_bddAbove_range_of_strictMono_of_common_denominator` are private Lean declarations: they are source-level audit markers, not importable global declaration names.

Mathematical assertion	**Formal correspondence and role**
Theorem 1.1: real closedness of $k((\Gamma))$	`hahnKaplansky_realClosed_of_realClosed_divisible`.
Theorem 1.2: real closedness of $\mathcal{P}(k)$	`puiseuxSeries_isRealClosed`.
Proposition 1.3: the ordered-field criterion for real closedness	`IsRealClosed.of_linearOrderedField`.
Lemma 2.1: leading-term normalization of a positive Hahn series	`HahnField.exists_leadingTerm_decomposition_of_pos`.
Lemma 2.2: binomial square root of a 1-unit	`HahnField.exists_square_root_of_orderTop_sub_one_pos`.
Proposition 2.3: nonnegative Hahn series are squares	`HahnField.isSquare_of_nonneg`.
Lemma 3.1: normalization at an odd-multiplicity Newton root	`HahnField.oddClusterHypotheses_scalePolynomial_comp_single_zero_of_rootMultiplicity`.
Definition 3.2: normalized odd Newton problem	`HahnField.KOddClusterHypotheses`.
Lemma 3.3: a nonzero odd-multiplicity root of an odd-degree polynomial	`exists_nonzero_root_odd_rootMultiplicity_le_natDegree`.
Proposition 3.4: multiplicity descent along a lower Newton edge	`HahnField.positiveHahnRoot_of_lowerNewtonEdge`.

Mathematical assertion	Formal correspondence and role
Definition 4.1: ordinal initial segments, successors, and limits	Standard mathlib concepts.
Lemma 4.2: successor-stage correction	`HahnField.exists_oddClusterStep_of_current_edge_initial_root`.
Lemma 4.3: strict increase of successive correction values	`HahnField.KOddClusterFixedStepChainFromData.stepValue_strictMono`.
Lemma 4.4: Hahn sum of a separated well-ordered family	`HahnField.HahnSeries.summableFamilyOfSeparatedSupports`.
Lemma 4.5: finite dependence of polynomial-evaluation coefficients	`HahnField.OrdinalSupport.HahnSummableFamily.eval_coeff_eq_predTrunc_of_support`.
Lemma 4.6: a root or a multiplicity-1 block successor	`HahnField.rootInMaximalIdeal_or_exists_blockSuccessor_at_state`.
Definition 4.7: coherent multiplicity-1 Newton block process	`HahnField.OneClusterCoherentBlockChain`.
Definition 4.8: coherent higher-multiplicity terminal-state process	`KSameMultiplicityOrdinalJetChain`.
Proposition 4.9: the multiplicity-1 limit prefix	`HahnField.PositiveIioCoherentChainAtBelow.limitStepOfNaturalComponents`.
Proposition 4.10: invariant of the higher-multiplicity terminal process	`iterateFixedLiftTerminalOrdinalPrefixState_invariant`.
Definition 4.11: the Hartogs ordinal	`hartogsOrdinal`.
Lemma 4.12: the Hartogs obstruction in Γ	`not_exists_strictMono_cardinal_succ_ord_toType`.
Lemma 4.13: resolution of a terminal obstruction	`HahnField.KSameMultiplicityFixedStepTerminalObstructionData.positiveHahnRoot_of_edgeStep_rootOrStep_terminalResolution`.
Proposition 4.14: solution of the multiplicity-1 branch	`kOddClusterRootAt_one_of_ordinalStackRec`.

Mathematical assertion	Formal correspondence and role
Proposition 4.15: termination of the higher-multiplicity branch	`positiveHahnRoot_of_sameMultiplicityZeroBranch`.
Lemma 5.1: a slope dominated by the top-degree term	`HahnField.exists_natDegree_dominating_slope`.
Theorem 5.2: roots of odd-degree polynomials in $k((\Gamma))$	`hahnKaplansky_exists_root_of_odd_natDegree`.
Definition 6.1: power-series identification of a fixed denominator level	`Puiseux.powerSeriesFixedDenominatorValuationSubringEquiv`.
Lemma 6.2: completeness of a fixed denominator level	`Puiseux.fixedDenominatorValuationSubring_isAdicComplete`.
Lemma 6.3: simple-root lifting at a fixed denominator	`Puiseux.levelSimpleRootLiftHypothesis_of_isAdicComplete`.
Lemma 6.4: denominator preservation at full multiplicity	`Puiseux.hasDenominator_newtonStep_of_full_initial_rootMultiplicity`.
Lemma 6.5: unboundedness of a strict chain in $\frac{1}{N}\mathbb{Z}$	`not_bddAbove_range_of_strictMono_of_common_denominator`.
Theorem 6.6: bounded-denominator roots of odd-degree polynomials	`puiseuxSeries_exists_root_of_odd_natDegree`.
Proposition 6.7: nonnegative Puiseux series are squares	`puiseuxSeries_isSquare_of_nonneg`.
Corollary 7.1: real closedness of $\mathbb{R}((\mathbb{Q}))$	`realRationalHahnField_isRealClosed`.
Corollary 7.2: real closedness of $\mathcal{P}(\mathbb{R})$	`realPuiseuxSeries_isRealClosed`.

B Reproducibility and references

The machine-checked developments can be rebuilt with:

```
lake build
lake build HahnKaplanskyRealClosedness.Examples
lake env lean HahnKaplanskyRealClosedness/AxiomAudit.lean
```

```
lake env lean HahnKaplanskyRealClosedness/Smoke.lean
lake env lean HahnKaplanskyRealClosedness/PuiseuxSmoke.lean
task lint
```

The public theorem audit uses only the standard foundational axioms:

`propext`, `Classical.choice`, `Quot.sound`.

References

[1] I. Kaplansky, *Maximal fields with valuations*, Duke Mathematical Journal 9 (1942), 303–321.

[2] A. J. Engler and A. Prestel, *Valued Fields*, Springer Monographs in Mathematics, Springer, 2005.

[3] R. J. Walker, *Algebraic Curves*, Princeton Mathematical Series 13, Princeton University Press, 1950.